

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Early Risk Stratification of Adult Emergency Department Patients Using Routinely Collected Triage Data

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Statement of the Problem:

Emergency departments need reliable ways to identify high-risk patients early, especially in busy triage settings where nurses must make rapid decisions using limited information. (De Freitas et al., 2020) At the first point of assessment, clinicians often depend on routinely collected data such as age, vital signs, GCS, oxygenation status and the presenting complaint. These data points can show early signs of deterioration but they may be difficult to interpret quickly when the department is crowded or staff are under pressure.(De Freitas et al., 2020; Tyler et al., 2024)

Artificial intelligence has the potential to support triage by helping identify patients who may need urgent review, admission or escalation of care. However, many AI-assisted triage tools are difficult to apply at the bedside because they rely on complex electronic health record data, are not tested in local workflows or do not clearly explain why a patient is being flagged. This is a problem because nurses need decision support that is clinically understandable, practical and safe to use during real triage.(Da'Costa et al., 2025; Tyler et al., 2024)

For Mercer General, the problem is not simply whether AI can predict risk. The larger issue is whether an AI-assisted tool can use routine triage data to support earlier recognition of high-risk adult ED patients while still keeping nurses and clinicians in control of the final decision.(Da'Costa et al., 2025)

Previous Work

Recent literature shows growing interest in the use of artificial intelligence and machine learning in emergency department triage. Tyler et al. reviewed how AI and machine learning have been applied in hospital ED triage and found that these tools may support patient prioritisation, earlier recognition of serious cases and decision-making around admission or urgent intervention. However, the review also highlighted the need for stronger clinical validation, safer implementation and careful attention to how these tools fit into real ED workflows.(Tyler et al., 2024)

Da'Costa et al. also reviewed AI-driven triage in emergency departments and found that AI could improve patient flow, reduce waiting times and support more consistent triage decisions. Their work is useful because it identifies the practical barriers that must be addressed before AI triage tools can be trusted. These include data quality, algorithmic bias, lack of explainability, ethical

concerns and clinician trust. These issues are directly relevant to Mercer General because a risk prediction tool would need to be understandable to nurses and safe enough to support clinical judgement rather than replace it.(Da'Costa et al., 2025)

Araouchi and Adda proposed TriageIntelli, an AI-assisted multimodal triage system that combines different types of patient information. This is relevant because ED triage is rarely based on one measurement alone. Nurses often interpret several data points together, including symptoms, vital signs and clinical indicators. Their work supports the idea that an AI triage tool should consider multiple routine triage variables rather than relying on a single vital sign. However, systems like this still require real-world validation to determine whether they are practical, safe and acceptable in clinical settings.(Araouchi & Adda, 2024)

Lin et al. developed an interpretable deep learning system that used triage information such as demographics, vital signs and chief complaints to predict triage level, hospitalisation and length of stay. This paper is especially relevant because it shows that early triage data can support clinically meaningful predictions. It also supports the explainability focus of this proposal because the authors recognised that clinicians are more likely to trust AI tools when the reasoning behind the output can be understood. However, their model was developed in a specific clinical setting, meaning it would still require local testing before being applied elsewhere.(Lin et al., 2024)

Sitthiprawiat et al. developed and internally validated an AI-based emergency triage model for predicting critical outcomes in ED patients. Their work is important because it focuses on outcomes that matter clinically, such as ICU admission and resource use, rather than only predicting a triage category. This supports the idea that AI-assisted triage should help identify patients at risk of serious clinical outcomes. However, internal validation does not prove that the model would work equally well in another hospital or patient population, so local validation remains necessary.(Sitthiprawiat et al., 2025)

De Freitas et al. explored patient flow in a Caribbean emergency department and identified practical issues affecting ED processes in the regional context. Although this paper is not an AI triage study, it is important for this proposal because it shows why local workflow matters. An AI tool designed for Mercer General should not simply copy models developed in other health systems. It should account for the realities of Caribbean ED patient flow, including bottlenecks, delays and resource limitations.(De Freitas et al., 2020)

Chang et al. studied the use of machine learning and natural language processing in ED triage to predict clinical disposition. This is relevant because it supports the use of both structured triage variables and unstructured information such as presenting complaints or triage notes. For Mercer General, this suggests that a future AI-assisted tool could begin with structured vital sign data but eventually include chief complaint text to improve risk identification. This paper strengthens the proposal by showing that routine triage information can be used in a more clinically meaningful way when structured and text-based data are combined.(Chang et al., 2024)

Overall, the literature suggests that AI-assisted triage has strong potential but also important limitations. Existing studies show that AI can support risk prediction, patient prioritisation and flow management. However, there are still gaps around explainability, local validation, early triage data use and Caribbean ED workflow relevance. These gaps support the need for a small, practical and explainable pilot at Mercer General.

Proposed Solution

My proposed solution is a 12-week pilot of a flow-aware, explainable triage risk support tool for adult emergency department patients at Mercer General. The tool would use routinely collected information from the early patient journey to support faster recognition of patients who may be at higher risk of deterioration, urgent clinical need or delay within the emergency department workflow.

The tool would not replace triage nurses or automate final triage decisions. Instead, it would support the triage process by helping staff collect relevant information earlier, identify concerning clinical patterns and explain why a patient may need prompt review. The final triage decision would remain with the triage nurse and clinical team.

A key feature of the proposed tool is that it would have three connected workflow layers.

Layer 1: Reception-assisted digital intake

In many emergency departments, the first interaction may happen before formal nurse triage. A receptionist or intake staff member could enter basic non-clinical information into the tool, such as age, sex or gender, arrival mode and presenting complaint. The intake screen could also include a structured red-flag checklist for obvious urgent symptoms such as chest pain, shortness of breath, fainting, severe bleeding, confusion or visible distress.

This step would not allow reception staff to make clinical decisions. Instead, it would help capture early information and prompt faster nurse review when a red-flag symptom is reported. For example, if a patient reports shortness of breath or chest pain at registration, the tool could display: "Priority nurse review recommended." The triage nurse would still be responsible for confirming the clinical concern, recording or validating vital signs and making the final triage decision.

Layer 2: Explainable clinical risk stratification

After the triage nurse records routine clinical observations, the tool would use routinely collected triage variables to assign the patient to a low, moderate or high-risk category. These variables could include:

- Age
- Gender
- Systolic blood pressure
- Diastolic blood pressure

Mean arterial pressure
Pulse
Temperature
Respiratory rate
GCS
SpO₂ where available
FiO₂ where available
Presenting complaint where available

The tool would not simply produce a risk score. It would also provide a short explanation in plain clinical language so that the nurse can understand why the patient was flagged. For example:

“High risk: low MAP and high pulse detected. Review for possible poor perfusion or clinical deterioration.”

This explainability is important because a nurse-facing tool must be understandable at the bedside. A risk category alone may not be enough if the user cannot see the clinical reasons behind it.

Layer 3: Flow-aware escalation support

The third layer would make the tool more specific to the realities of emergency department workflow. It would consider whether a patient who appears clinically high risk may also be vulnerable to delays in the ED system. For example, a patient who is high risk and likely to require oxygen support, imaging, admission, senior clinical review or close monitoring may need earlier escalation to avoid deterioration while waiting.

This layer is important because emergency department risk is not only about the patient’s vital signs. It is also about whether the patient may become delayed at a point in the care pathway. In this way, the tool would ask two practical questions:

1. Is this patient clinically high risk?
2. Is this high-risk patient also likely to be affected by delays in the ED workflow?

This three-part thinking makes the proposal more useful for Mercer General because it connects clinical risk recognition with patient flow awareness.

The pilot could be structured as follows:

Timeline	Deliverable
Weeks 1 to 2	Confirm the clinical problem, review the current triage and registration workflow and identify what information is collected at each stage.

Weeks 3 to 4	Prepare and clean the triage dataset, including checking missing values, validating vital sign ranges and documenting available variables.
Weeks 5 to 6	Build a first prototype using routine triage variables and simple risk rules or a basic machine learning model.
Weeks 7 to 8	Add the explanation layer so that each risk output includes clear clinical reasons for the flag.
Weeks 9 to 10	Test the tool on retrospective triage data to check whether the risk categories and explanations make clinical sense.
Weeks 11 to 12	Collect feedback from nurses or clinical reviewers on whether the tool is clear, useful and realistic for triage and refine the prototype based on that feedback.

This proposal is feasible because it uses information already collected during the early ED journey and keeps the nurse at the centre of decision-making. It also addresses the main gaps identified in the literature: the need for explainable nurse-facing decision support, practical testing using early triage data and better alignment with local patient flow challenges.

References

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